Reg. No. | | | | | | | | | | |

[14EC82]

EIGHTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

LOW POWER VLSI DESIGN

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing atleast two questions from each part.
2. Missing data may be suitably assumed.

PART - A

1. a. Discuss how power dissipation attributes influences at different levels of overall design process. **07**

   b. Explain energy bands in an unbiased MIS structure. **03**

   c. Write about gate-induced drain leakage concept. **04**

2. a. Derive an expression for dynamic power dissipation and energy per transition in VLSI circuits, taking into account the charging and discharging of capacitance in CMOS circuits. **10**

   b. List the different design aspects related to principles of Low power design. **04**

3. a. Discuss how parallel or pipeline architecture reduces supply voltage in case of architectural transformation. **07**

   b. Explain the concept of power optimization using operation reduction and operation substitution. **07**

4. a. Make use of pre-computation architecture discuss how optimization is achieved with example. **08**

   b. Discuss how transistor sizing contribute in optimization technique. **06**

PART - B

5. a. List the salient features of NMOS and Pseudo NMOS logic family. **06**

   b. With neat circuit diagram discuss pre-charge high DCSL and improved pre-charge low DCSL. **08**

6. a. List the different ways that multiple voltage design may affect IC layout **06**

   b. List and explain major challenges in deep sub-micrometer device and circuits. Also discuss how short channel effect can be minimized. **08**

7. a. Explain the operation of adiabatic dynamic logic inverter circuits. **06**

   b. Discuss different types of clock generation schemes. **08**

8. a. Explain different types of software power estimation. **07**

   b. Discuss how algorithm transformations are done to match computational resources. **07**

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